

Machine Learning & Deep Neural Architecture

Notes — Page 1

1. Transformer Architecture & Self-Attention

Transformers rely on Multi-Head Self-Attention mechanisms to process input sequences in parallel, computing dynamic attention weights between all token pairs using Query (Q), Key (K), and Value (V) projections:

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) * V$.

2. Positional Encodings & Feed-Forward Networks

- Rotary Position Embeddings (RoPE): Encodes relative token positions by rotating query/key vectors in complex space.
- Gated Linear Units (SwiGLU): Improves non-linear feature representation in transformer feedforward layers.

Machine Learning & Deep Neural Architecture

Notes — Page 2

3. Parameter-Efficient Fine-Tuning (PEFT)

LoRA (Low-Rank Adaptation) freezes pre-trained model weights and injects trainable rank-decomposition matrices into transformer layers, drastically reducing trainable parameters.

4. Quantization Techniques

- 4-bit NormalFloat (NF4): Optimized 4-bit data type for normally distributed neural weights.
- Double Quantization: Quantizes quantization constants to save additional memory.